Insurance Risk Analytics: Business Problem Statement

1. Business Background

The insurance company provides healthcare insurance services to employees across multiple corporate organizations. Through partnerships with companies from different industries, the organization offers medical coverage plans designed to support employee wellbeing while generating sustainable premium revenue for the business. Over the years, the company has experienced significant growth in the number of insured employees, corporate clients, and healthcare claims processed across its branches.

As the organization continues to expand, management has become increasingly concerned about the rising cost of claims, inconsistent customer claim behavior, and the long-term sustainability of the business. While some organizations contribute stable premium revenue with manageable claim expenses, others generate disproportionately high healthcare claims that place financial pressure on the company’s reserves and profitability.

2. Problem Statement

Currently, the organization does not have a standardized Risk Score system for evaluating the risk level of customers, employees, or corporate organizations. As a result, the company struggles to proactively identify high-risk customers before expensive claims occur. This limitation affects premium pricing decisions, profitability forecasting, fraud detection efforts, and long-term financial planning.

To address this challenge, management wants to leverage data analytics and business intelligence to develop a custom Risk Score model using the available insurance and company datasets. The goal is to calculate customer and organizational risk levels based on assumptions derived from factors such as:

- Age, BMI, and smoking habits
- Number of dependents and salary levels
- Claim frequency and claim amount
- Other employee-related attributes available within the datasets

3. Key Risk Assumptions

The company believes that the following factors are likely to drive higher risk and claim costs:

- Customers with higher BMI values may present higher medical risk
- Smokers may generate more expensive healthcare claims

- Older employees may require more medical attention
- Customers with frequent claims may represent higher insurance exposure
- Organizations with high claim-to-premium ratios may pose greater financial risk to the company

4. Business Objectives

Management intends to use the Risk Score model to achieve the following objectives:

- Improve underwriting decisions and optimize premium pricing
- Identify high-risk organizations and reduce claim losses
- Evaluate the profitability of corporate clients and insurance products
- Support strategic investment planning for premium revenues
- Identify suspicious or fraudulent claim patterns
- Improve operational efficiency and branch performance monitoring
- Support strategic investment planning to maintain sufficient reserves for future claims

5. Key Business Questions

The key business questions management seeks to answer include:

1. Which customers and organizations are classified as high risk based on the calculated Risk Score?
2. What factors contribute most to increased healthcare claim costs?
3. Which corporate organizations generate the highest financial exposure for the company?
4. Are premium payments sufficient to cover claim expenses across different organizations and insurance products?
5. Which customers or organizations generate claims that exceed their premium contributions?
6. How effectively can Risk Scores improve premium pricing and underwriting decisions?
7. Which branches perform best in terms of profitability, customer retention, and operational efficiency?
8. Are there suspicious or abnormal claim patterns that may indicate fraudulent activity?
9. Which demographic or lifestyle factors contribute most to high medical claims?
10. How can the company optimize investment strategies for premium revenues to ensure long-term sustainability?
11. What operational bottlenecks affect claims processing efficiency?
12. How can predictive analytics help forecast future claims, customer risk, and business profitability?

6. Expected Outcome

Through this analysis, the company aims to transition from reactive decision-making to a proactive, data-driven insurance management strategy that improves profitability, strengthens risk management, enhances operational efficiency, and ensures long-term business sustainability.

Using the datasets, the analytics team is expected to generate actionable insights that will support data-driven decision-making across the organization.